

When Do Hallucination Improvements Compose?

A Survey of Detection, Evaluation, and Mitigation in LLMs

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Abstract

Hallucinations undermine the reliable use of large language models (LLMs), motivating methods that detect errors, evaluate factuality, and intervene in generation. Yet improvements in these components do not automatically establish that a combined system produces more reliable, useful answers. This survey examines the compatibility of their evidence. We first distinguish world factuality from source faithfulness and organize detection and mitigation by the information they use and the stage at which they operate. We then review benchmark targets and conduct a source-linked documentary audit of representative methods. The comparison shows why reported safeguards must be interpreted at their tested scope: fact counts do not measure required information, separate verification prompts do not establish independent judgment, and component evaluations need not characterize errors after intervention. We synthesize these distinctions into compatibility conditions covering reference frames, information retention, feedback–evaluation dependence, and residual-error validation, with computational access as a practical constraint. Finally, we outline prospective tests for detector–intervention combinations and identify the limits of the available evidence. The accompanying materials provide the extraction protocol and source-linked evidence records.

1 Introduction

A fluent answer from a large language model (LLM) is not necessarily justified. In question answering (QA), a model may repeat a misconception; in summarization, it may add a plausible detail absent from the source. Both are called hallucinations, although their correctness criteria differ: source-unsupported content can be factually correct (Lin et al., 2022; Cao et al., 2022). Collapsing these settings obscures what a system improves.



Figure 1: Illustrative hallucination examples in source-grounded question answering and summarization. Green, pink, and blue denote unsupported additions, attribute errors, and relation errors, respectively. Unsupported content is not necessarily false.

In Figure 1, the library record supports 12,000 books and a second-floor archive. The cafe assertion lacks support; the changed count and location contradict the record. The delivery summary also invents a university source and reverses events. Verification must specify the claim and its reference evidence.

Better hallucination detection and better generation are often treated as complementary safeguards. Their published results can nevertheless concern different reference evidence, answer populations, and information budgets. A detector evaluated on unrevised answers may encounter a different error distribution after correction. Likewise, a higher fraction of supported claims can coexist with fewer useful claims. The central question is therefore *when evidence of a component improvement supports a claim about the combined system*.

We make three contributions. First, we connect reference frames, detection evidence, and intervention stages in a common comparison framework. Second, a bounded documentary audit records the tested safeguards and inference boundaries of rep-

representative methods, with primary-source locators. Third, we derive four compatibility conditions and a prospective reporting protocol from these comparisons. These are analytical contributions; we introduce no new detector or generation benchmark. Three questions organize the survey: **RQ1**, what reference defines an error; **RQ2**, what evidence supports a detector’s use; and **RQ3**, when can detection and mitigation evidence support reliable, useful generation?

Scope and organization. We cover text QA, long-form generation, summarization, dialogue, and retrieval-augmented generation (RAG), emphasizing ACL-family research and mechanism-defining work from other venues. Multimodal perception and general reasoning are excluded. Figure 2 maps concepts (§2), detection (§3), evaluation (§4), mitigation (§5), and analysis (§6).

Audit design. We fixed all thirteen named studies in an existing representative-method resource table before coding reporting properties. This purposive set spans source alignment, sampling, internal detection, training, retrieval, decoding, editing, revision, and abstention. For each primary full text, we inspected methods, evaluation, results, and relevant appendices, recording the reference frame, scoring unit, access, information control, feedback–evaluation separation, resources, and joint validation. A statement that evidence was not located is restricted to the inspected sections. All selected cases are retained; no field-wide frequency or pooled effect is inferred. Appendix A and the accompanying codebook distinguish this audit from the broader contextual literature search and document reading depth.

Relation to prior surveys. Ji et al. (2023) distinguish source faithfulness from world factuality across NLG tasks, while Wang et al. (2024) organize LLM factuality evaluation and enhancement and identify obstacles to fair comparison. Zhang et al. (2025b) and Alansari and Luqman (2026) analyze benchmarks, mitigation stages, evaluator limitations, and task-dependent trade-offs. Schmidtova et al. (2025) audit human-evaluation reporting, including definitions, annotation procedures, and agreement. Building on these perspectives, our bounded audit examines how representative studies document the conditions needed to compare or combine detection and mitigation. Appendix A.5 records the inspected overlap and limits on nov-

elty.

2 Concepts and Scope

2.1 Reference frames

World factuality concerns agreement with externally established facts; *source faithfulness* concerns support by supplied evidence. Neither entails the other: a response can repeat a false source faithfully or add a true but unlicensed detail (Cao et al., 2022; Adlakha et al., 2024). BEGIN evaluates attribution in grounded dialogue under this distinction (Dziri et al., 2022b).

Source-based labels should distinguish *supported*, *contradicted*, and *not established*, retaining the last separately from world falsity. World verification additionally needs a specified evidence source, date, and unresolved-claim policy. These are operational choices, not a universal definition. HalluLens revisits definitions and benchmark construction (Bang et al., 2025); comparisons must map study-specific labels before comparing scores.

2.2 Error relations and development

Entity, attribute, and relation errors are orthogonal to reference frames: a wrong date can contradict either a document or a world fact. Annotation units also differ. ANAH provides sentence-level analysis, while HalluEval 2.0 extends empirical factuality analysis across domains (Ji et al., 2024; Li et al., 2024).

Research moved from distinct task-specific correctness targets in 2022 toward atomic verification and black-box self-checking in 2023 (Min et al., 2023; Manakul et al., 2023), followed by contextual monitoring and factuality alignment. Recent studies scrutinize judge reliability, information coverage, and temporal validity (Godbole and Jia, 2025; Samarinas et al., 2025; Jiang et al., 2026).

Three axes organize the comparison: reference frame, detection evidence, and intervention point. Benchmarks evaluate generated content or detectors; detectors can guide decoding. Fine-tuning and long-context studies motivate failures in supervision, knowledge access, and evidence use (Gekhman et al., 2024; Liu et al., 2024c). Across axes, we compare evidence access, generator signals, supervision, and cost, separating decision criteria from deployment requirements.

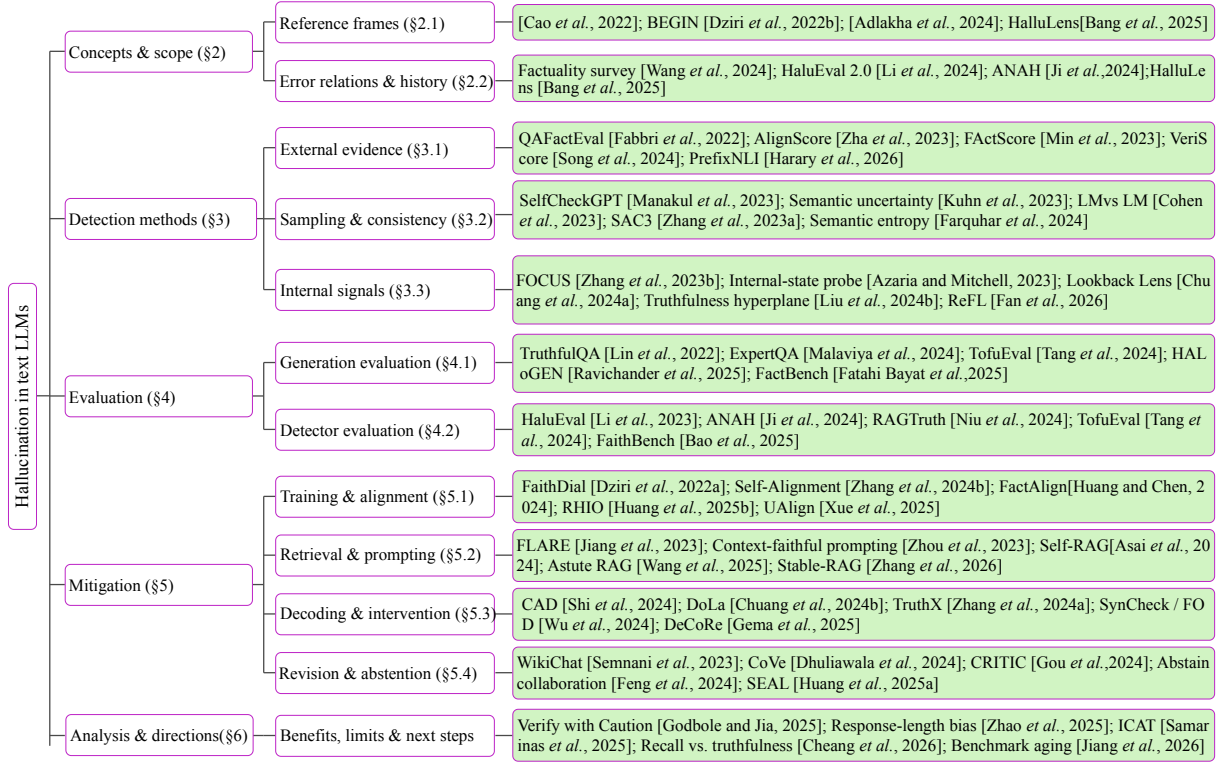


Figure 2: The structure of this survey. Green boxes list representative studies for each branch.

3 Detection: Evidence and Access

Detection estimates errors under a specified reference; Figure 2 groups methods by evidence and model access.

3.1 External evidence

Evidence-based methods compare outputs with supplied or retrieved knowledge. QAFactEval asks questions about summary content and checks answer recovery from the source; question generation and answerability affect its decisions (Fabbri et al., 2022). AlignScore learns alignment across heterogeneous tasks and aggregates source-chunk/output-sentence scores (Zha et al., 2023). Both use trained auxiliaries; neither guarantees correct verification.

FActScore decomposes long answers into atomic facts and estimates their supported fraction relative to a designated source (Min et al., 2023). VeriScore instead extracts verifiable claims, using surrounding context to resolve references (Song et al., 2024). Extraction changes the score denominator; missing support differs from contradiction. PrefixNLI extends entailment to incomplete prefixes, enabling feedback during decoding (Harary et al., 2026).

3.2 Sampling and consistency

SelfCheckGPT scores a response sentence against additional responses to the same prompt (Manakul et al., 2023). Its NLI and prompting variants aggregate contradiction or non-support judgments across samples. This requires repeated generation rather than target-model probabilities; agreement can still reflect a repeated misconception.

Semantic uncertainty instead clusters sampled answers by semantic equivalence and estimates entropy over meanings, discounting paraphrase variation (Kuhn et al., 2023; Farquhar et al., 2024). Sentence-level QA uses mutual entailment; FactualBio relaxes this to non-contradiction with at least one-way entailment. The weighted estimator needs sequence probabilities; a discrete approximation uses cluster frequencies. SelfCheckGPT localizes inconsistency in a given response, whereas semantic entropy measures dispersion among answer meanings, targeting unstable confabulations. Both depend on sampling and comparison quality; stable false beliefs need not trigger either signal.

3.3 Internal signals

FOCUS combines token probabilities with attention-based uncertainty propagation from an accessible proxy model (Zhang et al., 2023b).

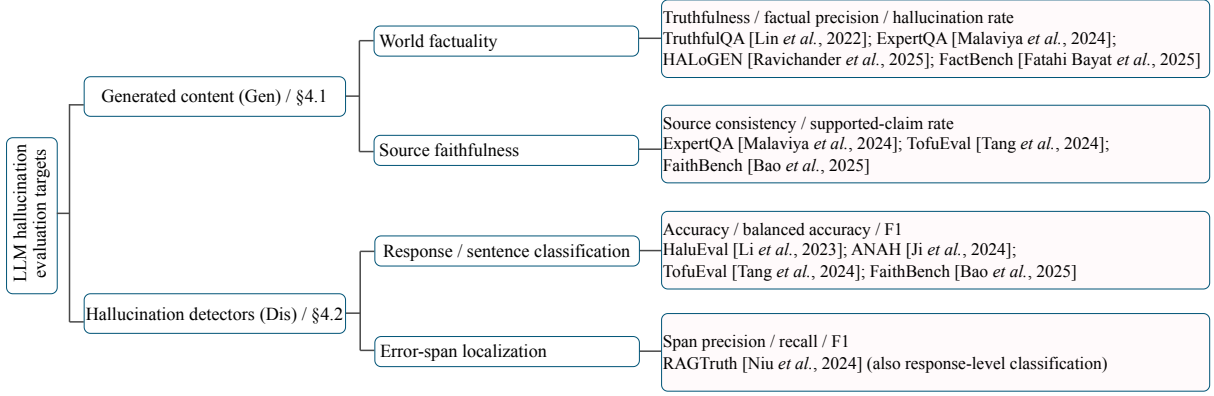


Figure 3: Taxonomy of hallucination evaluation methods for LLMs. The two branches distinguish generated-content evaluation from error-label prediction.

The target generator can remain black-box; the proxy must expose both signals. Hidden-state classifiers learn truth labels from activations (Azaria and Mitchell, 2023). Lookback Lens uses context-versus-generation attention features with a supervised linear classifier (Chuang et al., 2024a). Probes require representative training labels.

A separable activation pattern need not represent truth: associated falsehoods can resemble factual recall in controlled experiments (Cheang et al., 2026). Probes therefore need evaluation across natural error types and generators.

Source alignment cannot establish source truth; sampling and probes supply no independent factual evidence. “Black-box” and “reference-free” differ because access to generators, proxies, and verifiers must be specified separately.

4 Benchmarks and Evaluation

Figure 3 separates generation evaluation from detector evaluation. World factuality and source faithfulness are reference frames for both; detector sub-branches distinguish scoring units. Table 1 sizes count dataset units, not difficulty.

4.1 Generation evaluation

TruthfulQA evaluates misconception-oriented answers for truthfulness and informativeness, with a separate multiple-choice setting (Lin et al., 2022). ExpertQA distinguishes correctness from attribution through expert questions and claim-level judgments (Malaviya et al., 2024). HALoGEN uses domain-specific verifiers; FactBench combines user-derived prompts with web verification (Ravichander et al., 2025; Fatahi Bayat et al., 2025). Their scores are not interchangeable because evidence and verifier scopes differ.

TofuEval studies topic-focused dialogue summaries, where factuality must coexist with relevance and completeness (Tang et al., 2024). Precision needs coverage and refusal reporting, with explicit empty/unverifiable-answer policies.

Aggregation also changes the target. For $n > 0$ extracted claims and s supported claims, VeriScore combines $P = s/n$ with $R_K = \min(s/K, 1)$ as $F_1@K = 2PR_K/(P + R_K)$ for $s > 0$, and zero otherwise (Song et al., 2024). Its domain-specific $K > 0$ is the median extracted-claim count across evaluated responses. Unlike FActScore’s basic support fraction, this rewards supported-claim quantity as well as precision. However, R_K is not coverage of the information required by a question: extra supported details can raise it while leaving important aspects unanswered. Comparisons should fix K and separately audit extraction, verification, and task-relevant coverage. FactBench’s undecidable units additionally require an explicit aggregation policy (Fatahi Bayat et al., 2025).

4.2 Detector evaluation

HaluEval mixes generated task-specific examples with human-annotated general queries; recognition accuracy need not transfer to unselected errors (Li et al., 2023). ANAH annotates sentence-level types, evidence, and corrections; RAGTruth adds naturally generated RAG responses with error spans (Ji et al., 2024; Niu et al., 2024). Response classification and span localization are different tasks: finding one error does not locate every unsupported claim.

Table 1’s corpora do not supply every detector input. On fixed RAGTruth outputs, alignment can use source–response pairs (Niu et al., 2024). Self-CheckGPT additionally needs prompt-conditioned

Benchmark	Setting	Size	Focus	Metrics
TruthfulQA (Lin et al., 2022)	Gen/MC	817 questions (2022)	World factuality	Gen: truthfulness; informativeness; MC1 accuracy; MC2 true-answer mass
HaluEval (Li et al., 2023)	Dis	35,000 samples	Mixed-task errors	Recognition accuracy
ANAH (Ji et al., 2024)	Dis	~12k sentences	QA support; contradiction	Type accuracy; F1
RAGTruth (Niu et al., 2024)	Dis	17,790 responses	RAG faithfulness	Response/span P, R, F1
ExpertQA (Malaviya et al., 2024)	Gen	2,177 questions	Correctness; attribution	Human factuality/support
TofuEval (Tang et al., 2024)	Dis/Gen	1,479 summaries*	Dialogue summaries	Balanced accuracy; error rate
HALoGEN (Ravichander et al., 2025)	Gen	10,923 prompts	Domain factuality	Atomic hallucination rate
FaithBench (Bao et al., 2025)	Dis/Gen	750 summaries	Summary faithfulness	Balanced accuracy; macro F1; conditional error rate
FactBench (Fatahi Bayat et al., 2025)	Gen	1,000 prompts	Verifiable claims	Hallucination score; precision

* Paper-reported retained count; the source arithmetic is inconsistent. HALoGEN’s count includes programming. FaithBench’s selected cases are not a prevalence sample. Dis/Gen indicates distinct possible uses, not a shared metric.

Table 1: Hallucination evaluation benchmarks for LLMs. ‘Dis’, ‘Gen’, and ‘MC’ denote error-label discrimination, generated-content evaluation, and multiple choice, respectively; P and R denote precision and recall. Sizes count the stated units. Appendix B details the evaluation protocols.

samples from a declared generator; Lookback Lens needs attention features and probe-training labels. Specify feature models and prefixes; regenerating answers changes the evaluated outputs. Sentence-to-token conversion adds localization assumptions. TofuEval’s sentence and summary decisions also require separate assessment (Tang et al., 2024).

FaithBench selects cases by detector disagreement, so its error frequency is conditional on selection (Bao et al., 2025). TRIVIA+ tests long contexts and noisy supervision (Chen et al., 2026). Accuracy weights classes by prevalence; balanced accuracy weights them equally. Precision/recall require a declared positive class; span F1 requires tokenization and matching rules. Fix thresholds on validation data and match splits, generators, and evidence. Similar evaluator accuracy can still conceal biased system-level estimates (Godbole and Jia, 2025). Independently audit useful generation after mitigation.

5 Mitigation: Interventions and Assumptions

Figure 4 organizes interventions by generation stage; Table 2 contrasts reported safeguards with their inference boundaries. Context-aware decoding (CAD) favors supplied evidence, whereas As-tute RAG reconciles conflicting knowledge (Shi et al., 2024; Wang et al., 2025). Adhering to a record and answering what is true can require different policies; missing, ignored, and unreliable

evidence motivate different interventions.

5.1 Training and alignment

FaithDial supplies curated source-grounded dialogue revisions for generation and critic supervision (Dziri et al., 2022a). Self-Alignment improves self-evaluation through Self-Knowledge Tuning, ranks candidate responses by factuality, and forms response-level preference pairs for direct preference optimization (DPO) (Zhang et al., 2024b). For long answers, atomic-claim scores are averaged before ranking, so training preferences remain response-level.

FactAlign combines response-level Kahneman–Tversky optimization (KTO) with sentence-level fine-grained KTO (fKTO) (Huang and Chen, 2024). Each sentence is treated as a continuation conditioned on the prompt and preceding sentences; binary factuality labels derive from support against retrieved Wikipedia evidence. Unlike DPO, fKTO needs no sentence-level preference pairs. UAlign incorporates confidence and semantic entropy into a reward model before proximal policy optimization (PPO) alignment (Xue et al., 2025). RHIO targets contextual faithfulness: masking retrieval heads produces negative examples, control tokens distinguish faithful from unfaithful continuations, and contrastive decoding subsequently separates them (Huang et al., 2025b). Finer feedback still inherits supervision errors.

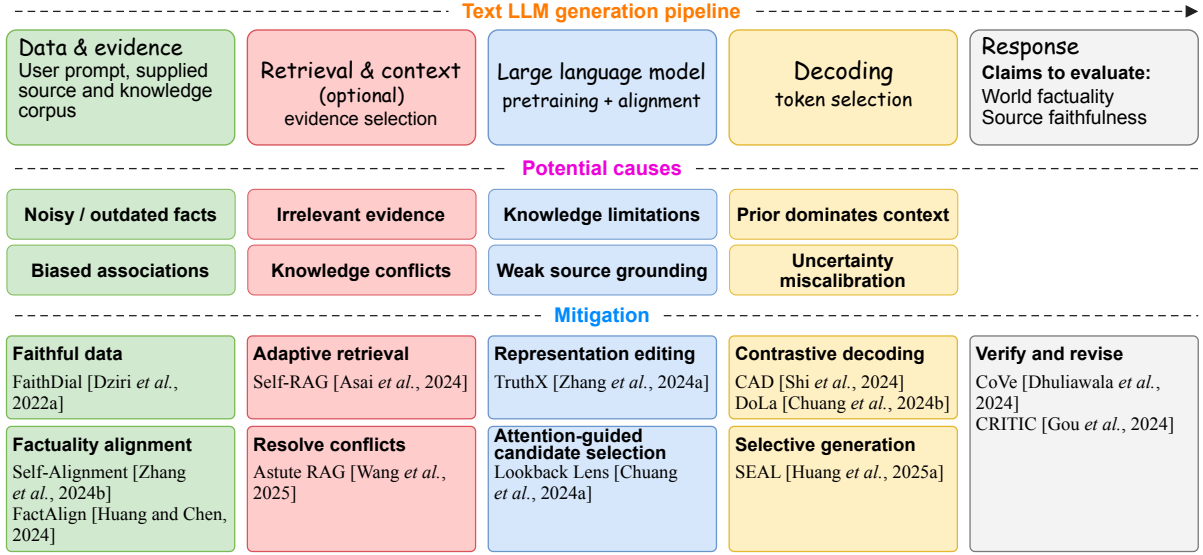


Figure 4: Potential causes and mitigation methods of hallucination in LLMs. Colors group stages of the generation pipeline; hybrid methods may span multiple stages.

5.2 Retrieval and prompting

FLARE retrieves actively when provisional continuations signal uncertainty, while context-faithful prompting changes instructions without retraining (Jiang et al., 2023; Zhou et al., 2023). Retrieval adds information but introduces selection errors and evidence conflicts. Self-RAG trains retrieval and reflection tokens that control when to retrieve and assess relevance, support, and utility; candidate continuations can then be ranked using these signals (Asai et al., 2024). Astute RAG instead uses a training-free, source-aware consolidation procedure: it elicits internal knowledge, compares it with retrieved passages, and iteratively resolves conflicts (Wang et al., 2025). Its reliability judgments remain fallible. These methods require separate audits of retrieval, conflict resolution, and evidence use.

5.3 Decoding and internal intervention

CAD amplifies the difference between context-conditioned and context-free token distributions (Shi et al., 2024). For model parameters θ , next token y_t , context c , query x , prefix $y_{<t}$, and contrast strength $\alpha \geq 0$,

$$p_{\text{CAD}}(y_t) \propto \frac{p_{\theta}(y_t \mid c, x, y_{<t})^{1+\alpha}}{p_{\theta}(y_t \mid x, y_{<t})^{\alpha}}. \quad (1)$$

CAD favors tokens whose probability increases when context is supplied, assuming that context deserves greater weight. DoLa instead subtracts earlier-layer from mature-layer log probabilities

within a plausible-token set, dynamically selecting an earlier candidate layer by maximum Jensen–Shannon divergence from the mature layer (Chuang et al., 2024b). It assumes layerwise changes reveal useful parametric knowledge. Both reshape token distributions without independently checking truth.

TruthX learns separate semantic and truth-related latent spaces and edits internal representations along a learned direction (Zhang et al., 2024a). Lookback Lens trains a classifier over context-versus-generation attention ratios and selects candidate continuations with that classifier (Chuang et al., 2024a). Its mitigation closes the detector–generator loop, but also inherits detector errors. Such steering does not establish universal truth representations; architectural transfer needs testing.

5.4 Revision and abstention

Chain-of-Verification (CoVe) drafts an answer, generates verification questions, and revises the draft; its factored variants answer verification questions without the original draft (Dhuliawala et al., 2024). Separating verification from the draft reduces answer contamination, but all stages can still share a parametric misconception. CRITIC adds tool feedback before revision, providing an external correction signal (Gou et al., 2024). WikiChat also grounds dialogue through retrieval and verification, coordinating these operations (Semnani et al., 2023). External feedback remains limited by search and tool reliability.

Case / source location	Reported safeguard or experiment	Boundary for comparison or composition
FActScore, §3.1, Tables 1/5, App. C	Support relative to a specified knowledge source; responding fraction and fact counts; editing using human factuality labels.	Information quantity is reported. Oracle-label editing does not validate the automatic estimator–editor combination.
CoVe, §3.3 / Table 3	Biographical FActScore and fact quantity are reported together; factored verification answers are generated without the original draft.	Precision and retained content can change together. Prompt separation does not establish independent factual knowledge.
Self-RAG, App. C.2	Human checks of outputs and reflection tokens on 50 examples; support/plausibility excludes self-predicted irrelevant/no-support cases.	Human checks supplement self-scoring, but the conditional support/plausibility estimate does not characterize all outputs.
DoLa, Table 1 / §4.2 / Apps. E–F	Truthfulness and informativeness; latency measurements under a specified device, batch, and generation length.	Measured overhead applies to that setup; it is not an equal-budget ranking against retrieval or revision methods.
SEAL, §6 / App. E	A separate rejection experiment complements generation tests; tested beam settings differ from greedy baselines.	Generation factuality, selective answering, and compute allocation must be distinguished.
CRITIC, App. D	Detection metrics on initial answers and human analysis complement tool-assisted correction results.	These component analyses do not by themselves quantify fixed-detector calibration on revised errors.

Table 2: Contrasts from the documentary audit. Locations refer to the cited primary papers discussed in Sections 5–6; the supplementary records retain all thirteen cases. A boundary restricts the inference drawn here, not the validity of the original study.

SEAL learns a rejection token and uses its probability to penalize uncertain decoding trajectories; the main experiments prohibit emitting that token, while a separate extension enables refusal (Huang et al., 2025a). Refusal and revision need coverage and information retention alongside precision.

6 Analysis and Future Directions

The audit in Table 2 yields four compatibility conditions. They are requirements for interpreting combined evidence, not a new scalar metric. We distinguish reported safeguards from stronger conclusions that those safeguards do not establish.

C1: Align the reference frame and evaluated population. FActScore operationalizes factual precision as support relative to a designated knowledge source, not unqualified world truth (Min et al., 2023, §3.1). CAD strengthens use of supplied context (Shi et al., 2024); DoLa and TruthX target truthfulness using internal signals (Chuang et al., 2024b; Zhang et al., 2024a). A faithful answer to a false document can therefore satisfy one target and violate another. Similarly, AlignScore’s source alignment, SelfCheckGPT’s sampled consistency, and semantic entropy’s meaning dispersion observe different evidence (Zha et al., 2023; Manakul et al., 2023; Farquhar et al., 2024). They cannot be ranked by treating all reported scores as interchangeable hallucination rates. A defensible comparison fixes the intended reference, the answer unit, and the evidence/unknown policy before matching metrics.

Evaluation on selected disagreements or accepted answers must retain its conditional denominator.

C2: Preserve the information needed by the task.

The audit does not support a blanket claim that methods ignore information retention. FActScore reports responding fractions and facts per response; DoLa reports truthfulness together with informativeness. Semantic entropy explicitly evaluates selective prediction through rejection-accuracy curves over retained-prediction fractions and their area (AURAC) (Farquhar et al., 2024). CoVe reports both factual precision and fact quantity: in its Llama 65B biography experiment, factor+revise increases FActScore from 55.9 to 71.4 over few-shot prompting while the mean number of facts decreases from 16.6 to 12.3 (Dhuliawala et al., 2024, Table 3). This documents a change in both dimensions; it does not isolate a causal length effect or establish lost task-relevant information.

The distinction is between *reporting a safeguard* and *controlling the desired quantity*. Fact counts, answer informativeness, and the fraction of questions answered are not required-aspect coverage. ICAT’s aspect-based evaluation and studies of response length sharpen this distinction (Samarinas et al., 2025; Zhao et al., 2025). SEAL further separates ordinary generation with rejection-aware decoding from an explicit rejection experiment (Huang et al., 2025a, §6). A comparison should report useful supported content and refusals, then compare risk at matched answered-question cover-

age where selection is involved.

C3: Trace feedback–evaluation dependence.

Independence has several meanings. CoVe separates verification answers from the initial draft, reducing draft contamination without introducing independent knowledge. FactAlign trains with source-derived factuality labels, while Self-RAG learns reflection judgments used in generation (Huang and Chen, 2024; Asai et al., 2024). Their evaluation must therefore be interpreted in relation to the feedback mechanism, not labeled independent merely because a separate stage or prompt exists.

Self-RAG provides a useful positive example and a boundary: Appendix C.2 reports human assessment of outputs and reflection tokens on 50 examples, but its support-and-plausibility assessment excludes cases predicted irrelevant or unsupported. Its support-and-plausibility estimate is conditional, rather than an all-output estimate. Lookback Lens uses GPT-4o for training labels and part of final judging, alongside human checks on XSum and NQ (Chuang et al., 2024a, App. B); the audit retains both the shared component and the additional verification. Evaluator accuracy alone can also hide class-specific bias in system-level scores (Godbole and Jia, 2025). Final assessment should disclose shared models, evidence, labels, and selection rules and audit both high- and low-scoring outputs.

C4: Validate the detector on errors after intervention.

A component test is informative without being a complete composition test. FActScore’s Appendix C editing experiment uses human factuality labels, deliberately isolating estimator errors (Min et al., 2023). CRITIC reports initial-answer detection and human error analysis alongside correction results (Gou et al., 2024, App. D). Lookback Lens explicitly uses its learned detector to select continuations (Chuang et al., 2024a). These are distinct designs; none should be collapsed into a binary claim that joint analysis is absent or sufficient.

A remaining question is whether a detector’s operating point transfers to the residual errors produced by an intervention. Reasoning prompts can change both error frequency and detection cues (Cheng et al., 2025); consistent falsehoods can evade sampling-based signals (Farquhar et al., 2024; Cheang et al., 2026). For erroneous candidate E , release decision R , and $P(E) > 0$, the

elementary identity

$$P(E \cap R) = P(E)P(R | E) \quad (2)$$

shows why reducing candidate error alone does not guarantee fewer erroneous releases. It assumes no independence and is not itself a new theoretical result. Prospective tests should compare baseline, detector-only, mitigation-only, and combined systems on fixed questions, report erroneous releases per query, conditional error among released answers, and coverage, and recalibrate only on held-out validation data.

Resource compatibility is an additional constraint.

Reported costs are not uniformly absent. DoLa measures latency in a fixed setup; SelfCheckGPT reports an API cost estimate. Such values cannot establish a hardware- and workload-independent ordering. Training-based FactAlign, retrieval-based Self-RAG, and iterative CoVe/CRITIC move expenditure between offline and online stages. SEAL’s tested beam search and greedy baselines also use different decoding budgets. Report access requirements, training/annotation expenditure, generated tokens, retrieval/tool calls, and measured latency separately. Appendix D makes these conditions an unperformed prospective protocol, rather than presenting the audit as evidence of a new system’s improvement.

7 Conclusion

RQ1 requires a declared reference frame, claim unit, and unresolved-label policy. For **RQ2**, evidence access determines which errors a detector can identify, while reported safeguards must be read at their tested scope. For **RQ3**, the audited cases support four compatibility checks: align the correctness target, retain useful information, separate optimization from final judgment, and re-evaluate detection on post-intervention errors. The survey does not rank heterogeneous methods or establish a universal performance gain. It provides traceable comparisons and a prospective protocol for testing whether component improvements translate into more reliable useful generation.

Limitations

This is a critical narrative survey with a purposive documentary audit. The audit sample was fixed from an existing representative-method table before the new coding round; it is neither random nor

an exhaustive sample of hallucination research. Its selection favors English text tasks and mechanism diversity. The broader discovery library and targeted update search have different reading depths from the audit; their records are not additional audited cases. Source-linked extraction is AI-assisted, not independent human double annotation. Ambiguous or unlocated evidence is retained as such. We report no agreement statistic, pooled effect estimate, model replication, or field-wide prevalence estimate. Heterogeneous tasks, evidence, judges, and budgets prevent a defensible global ranking. The compatibility conditions are a synthesis, not a validated predictive model; their operational benefit requires prospective testing. Figure 2 is a representative map rather than an exhaustive bibliography.

Ethical Considerations and AI Assistance

Mischaracterizing a reliability safeguard as a guarantee can encourage unsafe deployment. We therefore distinguish original-study observations, our interpretation, and unperformed tests. The work analyzes public research documents and introduces no human-participant study or trained model. Full papers are linked rather than redistributed. Visual-design sources are credited in Appendix A; illustrative interactions are synthetic. OpenAI Codex assisted literature retrieval, extraction, analysis, drafting, and artifact preparation. AI-assisted cross-checking does not constitute independent human verification. The evidence records expose source locations and uncertainties for author and reader inspection; responsibility for the submitted claims remains with the human author.

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A Selection, Provenance, and Reading Depth

A.1 Two evidence layers

The contextual library supports a narrative account; the documentary audit supplies the comparative cases in Section 6. Their populations and reading depths must not be conflated. The audit includes AlignScore, FActScore, SelfCheckGPT, semantic entropy, Lookback Lens, FactAlign, Self-RAG, CAD, DoLa, TruthX, CoVe, CRITIC, and SEAL. These were all the named studies in a pre-existing resource table, chosen for mechanism diversity. The sample was fixed before the new extraction round on 5 October 2026; this is a recorded protocol, not an externally preregistered study.

We recorded primary publication URLs, PDF digests, inspected sections, and claim-specific locators. Extraction fields cover reference frame, scoring unit, evidence/access, information control, feedback–evaluation separation, resources, and joint validation. Information controls are described rather than collapsed into an adequate/inadequate score: fact quantity, task quality, required-aspect coverage, and answered-question coverage measure different things. Different prompts or model names do not prove statistically independent errors. A qualitative access requirement is not measured latency. “Not located” is bounded by the inspected source passages and is not proof of absence. The supplementary CSV preserves caveats for every case. Source checks and revisions were AI-assisted; no independent human annotation or inter-annotator agreement is claimed.

A.2 Contextual discovery and targeted update

The original contextual metadata snapshot is 15 September 2026. Discovery queried official ACL Anthology collections across ACL, EMNLP, NAACL, EACL, Findings, and TACL in 2022–2026. Twenty-five collections were available and five requested combinations unavailable. A broad title filter returned 1,766 records; three targeted additions and seven external studies brought the candidate pool to 1,776. The 87-study working library was selected for relevance, mechanism diversity, benchmark coverage, and counterevidence. Automatic title triage excluded 401 candidates; 1,288 remained unassessed reserve records. All selected titles and abstracts were screened; initial preparation inspected passages in fifteen papers, followed by claim-specific checks. These are workflow counts,

not a full-text systematic-review flow.

The targeted update on 5 October 2026 checked closely related surveys and evaluation work against primary publication records. Its query and inclusion log is supplied separately; it does not purport to exhaust all intervening publications. Canonical published versions are preferred. The audit sample remains fixed, so update records and the broader bibliography do not increase its denominator. No model experiments, independent benchmark replications, or pooled effect estimates were performed.

A.3 Visual provenance and coding examples

The layouts of Figures 1, 3, and 4 and Table 1 follow Liu et al. (2024a); Figure 2 follows Liu et al. (2025). Their content addresses text LLMs. Figure 1 uses synthetic interactions. Figure 4 summarizes mechanisms proposed in the literature, not an experimentally identified causal decomposition. The figures provide a representative conceptual map; the source-linked audit adds protocol details that cannot be inferred from this map.

Detection branches are assigned by decision evidence, mitigation branches by the operation that changes generation. Thus Lookback Lens appears as attention-based detection and candidate-selection mitigation. RHIO combines training on induced negatives with contrastive decoding. Hybrid methods need not belong to exclusive categories. Source faithfulness is recorded separately from world factuality, and target-generator access separately from proxy or verifier access.

A.4 Compatibility audit records

The accompanying CSV contains one record per fixed case, with primary-source URLs, PDF digests, inspected sections, and locators for each substantive comparison. The records describe concrete reported controls rather than assign a universal quality score. They distinguish measured costs from access requirements, human labels from model judgments, and component analyses from tests of the complete detector–intervention system. Table 2 highlights contrasts used in the main argument. Other selected studies remain in the record even when they do not support a simple positive or negative code. Tables 3–5 summarize reported information controls and their boundaries.

Study	Reported information control	Interpretation boundary
AlignScore (Zha et al., 2023)	Fixed-output metric evaluation. Explicitly separates omission/content-selection errors from factual-consistency errors in PolyTope coding. A required-aspect-completeness or selective-answering outcome measure was not located in the inspected evaluation sections; omission exclusion is not such a measure.	Factuality and information selection are deliberately separated; detector performance does not alone establish usefulness after applying an output intervention. The altered PolyTope error definition and dataset overlap exclusions must be retained in comparisons.
FActScore (Min et al., 2023)	Reports responding fraction and fact quantity per response; explicitly warns that precision does not measure recall. The related editing experiment measures token preservation and agreement with human edits. Fact quantity and edit similarity are not required-aspect completeness.	Unsupported includes source absence as well as contradiction. Responding fraction and fact counts partly expose selective/short responses but cannot establish complete answers. Oracle-label editor results must not be attributed to the automatic estimator.
SelfCheckGPT (Manakul et al., 2023)	Fixed-output detection/ranking on generated biographies. Descriptive token/sentence statistics are reported, but matched answered-question retention or required-aspect-completeness evaluation was not located in inspected §§6–7 and Appendix D.	Sample consistency can miss stable incorrect beliefs; the observed task is a selected GPT-3 biography set. Sentence PR-AUC and passage correlation do not measure correctness/completeness after editing or abstention.
Semantic entropy (Farquhar et al., 2024)	Explicit answered-question retention through rejection-accuracy curves and AURAC; FactualBio also evaluates retained fact predictions. This controls a retained-unit fraction, not required-aspect completeness of long answers. Biography question generation attempts to cover claim aspects but has acknowledged misses.	QA labels operationalize reference-answer agreement, not independently established world truth; SI explicitly preserves that target when a reference is wrong. FactualBio has only 21 biographies/150 claims. Retained-unit coverage differs from semantic completeness; systematic low-entropy errors remain outside the targeted confabulation mechanism.
Lookback Lens (Chuang et al., 2024a)	Reports a general task-quality proxy (original MT-Bench score) alongside contextual-faithfulness score and NQ exact match. These do not establish matched required-aspect coverage or answer-completeness preservation for XSum.	The quality safeguard is task-specific and is not an equivalence test or matched-completeness study. Shared GPT-4o labeling/judging can preserve common errors despite human audit. Resource figures are approximate study-specific observations; transfer needs an auxiliary fitted head mapping.

Table 3: Documentary audit of reported controls (1/3). Descriptions retain original-study targets and unresolved reporting; precise passage locators and source digests are in the accompanying evidence records.

Study	Reported information control	Interpretation boundary
FactAlign (Huang and Chen, 2024)	Fact counts, factual precision and f1 @K; literal Section 3.1 recall term is min(1, number of generated atomic statements/K), not a reference-aspect recall. MT-Bench provides helpfulness proxy. No matched answered-question coverage comparison located in inspected sections.	Corpus support is not universal truth. Generated fact quantity is not required-information completeness. For loss ablation compare the one-iteration full model with its ablation, not the three-iteration headline result.
Self-RAG (Asai et al., 2024)	ASQA correctness, ROUGE, MAUVE and citation precision/recall; biography factual precision; ASQA maximum 100 generated tokens across baselines to limit length bias. Weight sweep exposes support-versus-fluency trade-off. These are not matched answered-question coverage or required-aspect completeness controls.	S&P denominator is conditional on the model support/relevance decisions. Reflection agreement with GPT-4 is not independent truth adjudication. Citation recall concerns generated claims, not all information a user requires.
CAD (Shi et al., 2024)	Summarization reports ROUGE-L alongside FactKB and BERT-precision; QA reports exact match. Task quality is monitored, but neither reference-aspect completeness nor matched answered-question coverage is established.	NQ-Swap intentionally changes source entities, so success may require contradicting world knowledge. Do not classify its exact-match gain as world-truth improvement or infer a twofold latency from two distributions.
DoLa (Chuang et al., 2024b)	TruthfulQA reports Truth, Info, their joint metric, and Reject; GPT-4 separately rates grammaticality/cohesiveness. These show informativeness/refusal trade-offs but are not matched coverage. For LLaMA-33B, Truth falls 62.5 to 56.4 while Info rises 69.0 to 92.4 and joint score rises 31.7 to 49.1.	Cannot add unlearned factual knowledge. Raw truth and informativeness may move in opposite directions. Latency result belongs to the reported hardware/output-length configuration.

Table 4: Documentary audit of reported controls (2/3). Descriptions retain original-study targets and unresolved reporting; precise passage locators and source digests are in the accompanying evidence records.

Study	Reported information control	Interpretation boundary
TruthX (Zhang et al., 2024a)	Truth, Info, joint score and no-comment counts; original-model perplexity and cross-task accuracy supplement quality. They monitor refusal/informativeness, not matched answered coverage or required information completeness.	Two-fold in-domain validation and multiple-choice transfer do not establish general free-generation truth calibration. Editing cannot guarantee truth or inject missing knowledge (Limitations p.10).
CoVe (Dhuliawala et al., 2024)	Correct/incorrect entity counts, multi-span precision/recall/F1, biography fact count with FActScore. Same-generator biography result 55.9 to 71.4 FActScore accompanies 16.6 to 12.3 facts. A separate Llama-2-chat clipping control is not a matched-length CoVe control.	Precision gain does not establish preservation of all correct or required facts. Closed-book checks remain bounded by model knowledge; shorter output is a measured trade-off, not proof of the entire gain being due to deletion.
CRITIC (Gou et al., 2024)	QA EM/F1 and qualitative/manual refusal and incorrect-correction categories; math answer correctness; toxicity task reports fluency/diversity proxies. No matched answered-question coverage or required-aspect preservation located. Non-oracle CRITIC and oracle CRITIC* must be kept separate.	Table 6 error-type percentages have a less explicit denominator than a full transition matrix; do not treat 10% incorrect correction as the population rate. Oracle CRITIC* is not deployable verification; math/toxicity results do not directly measure factual hallucination.
SEAL (Huang et al., 2025a)	Biography correct-claim counts and FActScore; LongFact precision/R@48/F1@48; separate refusal experiment reports known-question accuracy versus unknown-question rejection across tau; instruction following is checked. R@48 uses truthful-fact count relative to 48, not required-aspect recall or matched question coverage.	Known/unknown status is a correctness/popularity proxy, not direct access to parametric knowledge. Main result cannot be interpreted as answer abstention. Biography sample count conflicts between Table 8 (138) and Appendix B.2 (183), unresolved.

Table 5: Documentary audit of reported controls (3/3). Descriptions retain original-study targets and unresolved reporting; precise passage locators and source digests are in the accompanying evidence records.

A.5 Comparison with prior surveys

The source-linked prior-survey record covers general NLG hallucination, LLM factuality, broad LLM hallucination, human-evaluation reporting, and detection/mitigation reviews. Its purpose is to delimit the contribution, not establish that a concept is absent from an entire literature. The resulting contribution claim is a cross-stage compatibility synthesis applied to a bounded sample. No first-survey or first-audit claim is made.

B Benchmark Units and Protocol Caveats

Table 6 supplies the counting and interpretation details behind Table 1. A benchmark may support both a description of generated errors and evaluation of a detector, but those uses require different denominators and sampling assumptions. In particular, reporting error frequencies within deliberately selected difficult examples does not estimate a generator’s error prevalence on ordinary requests.

C Supplementary Comparative Evidence

Tables 7–10 extend the main synthesis with complementary studies. The final column states the inference boundary used in this survey; it is not an assertion that the original authors omitted every proposed test. These studies expand coverage of mechanisms and failure conditions rather than form a ranked list. Descriptive entries are grounded in the recorded primary metadata and selected source passages, with reading depth retained in the repository.

Benchmark	Unit, protocol, and limitation
TruthfulQA	The original 2022 benchmark contains 817 questions. Generation measures truthfulness/informativeness; MC1 measures single-true accuracy, and MC2 normalized probability mass assigned to true answers. The official January 2025 update introduced two-option choices and revised the question set. Report the dataset version and MC protocol separately; answer selection is not detector classification.
HaluEval	The 35,000 examples combine task-specific generated examples with a human-annotated general-query portion. Artificial and naturally generated errors should be reported separately rather than treated as one deployment distribution.
ANAH	Approximately 12,000 annotated sentences provide error types, evidence, and corrections. Generative versus discriminative annotators describe detector implementations; their task remains prediction of annotation labels.
RAGTruth	The 17,790 responses come from 2,965 prompts and six generators, not 17,790 independent documents. Response classification and span localization have different matching rules and must be evaluated separately.
ExpertQA	The 2,177 expert-curated questions support factuality and attribution analysis. Correctness and cited-source support are separate judgments; exact evaluation subsets and answer-generation conditions matter.
TofuEval	Section 3.3 reports 1,500 initial summaries, removal of 23, and 1,479 retained summaries. Those values are arithmetically inconsistent. We retain the reported 1,479 rather than invent a corrected dataset size. Topic relevance and completeness are separate from factual consistency.
HALoGEN	The 10,923 prompts cover all nine domains, including programming. That is the complete-suite count, not a verified text-only subset size. Different domain verifiers do not constitute one universal human truth oracle.
FaithBench	The 750 summaries derive from 75 passages and ten models, selected using detector disagreement. This supports difficult-case detector analysis; error frequencies are conditional on selection, not population prevalence.
FactBench	The 1,000-prompt snapshot uses user-derived prompts and fine-grained verification. Supported, unsupported, and undecidable units require explicit aggregation rules. A dynamic benchmark needs a fixed version for reproducible comparisons.

Table 6: Evaluation units, protocols, and limitations of the benchmarks in Table 1.

Study	Mechanism or evaluation design	Comparative implication and boundary
TRUE (Honovich et al., 2022)	Standardizes factual-consistency datasets across tasks and evaluates metrics at the example level.	Instance discrimination differs from system-level correlation. Combined existing tasks do not estimate real-world hallucination prevalence.
AggreFact analysis (Tang et al., 2023)	Stratifies summarization error annotations by generating model and error type; metric gains vary across strata.	Pooled scores can hide weaknesses on newer generators or rare error types.
BUMP (Ma et al., 2023)	Uses minimally differing human-written summary pairs with one introduced error.	A controlled error-sensitivity probe complements natural generation data; minimal pairs do not reproduce the full distribution of model errors.
ALCE (Gao et al., 2023)	Evaluates answer quality and citation quality in end-to-end retrieval and cited generation.	Citation presence, citation support, and factual correctness are distinct criteria. A plausible citation is not by itself evidence.
LongEval (Krishna et al., 2023)	Studies fine-grained human judgments and partial annotation for long summaries.	Fine-grained annotation requires agreement and coverage reporting; partial annotation does not exhaustively verify all claims.
FACTOR (Muhlgay et al., 2024)	Transforms factual corpora into true versus false continuation choices.	Controlled factual choice evaluates a different task from open-ended generation and error localization.
ReEval (Yu et al., 2024)	Perturbs evidence to create dynamic tests of evidence use under retrieval.	Static accuracy may reflect memorized knowledge rather than source adherence; counterfactual evidence requires an explicit faithfulness target.
Core (Jiang et al., 2025)	Filters redundant or uninformative subclaims before factual precision aggregation.	Trivial repetitions can inflate precision; informativeness and coverage are complementary criteria.
TRIVIA+ (Chen et al., 2026)	Adds long-context RAG detection data and controlled sample-dependent label noise.	Long-context and noisy-label tests complement evaluation on clean labels.
RGB (Chen et al., 2024)	Separates noise robustness, negative rejection, information integration, and counterfactual robustness in English/Chinese RAG.	Retrieval conditions define distinct RAG abilities, not one interchangeable hallucination score.

Table 7: Additional benchmarks and evaluation protocols for hallucination in LLMs.

Study	Mechanism or evaluation design	Comparative implication and boundary
WeCheck (Wu et al., 2023)	Aggregates weak labels over actual generated samples, then learns a noise-aware classifier.	Supervision quality and error distribution distinguish a learned consistency model from zero-shot entailment.
FactKB (Feng et al., 2023)	Uses knowledge-base factual pretraining with entity/relation objectives for factuality evaluation.	Structured supervision targets relation errors; broader transfer requires separate evidence.
TrueTeacher (Gekhman et al., 2023)	Uses LLM-labeled model-generated summaries to train a smaller consistency evaluator.	Distillation trades annotation cost for teacher-label dependence; generated errors differ from artificial perturbations.
FENICE (Scirè et al., 2024)	Extracts atomic claims and aligns them to a source using natural language inference (NLI).	Local alignment improves interpretability, while claim extraction and entailment both remain possible error sources.
FactReasoner (Marinescu et al., 2025)	Models evidence–claim logical relationships probabilistically after decomposition/retrieval.	Joint reasoning differs from independently verifying claims; its result still depends on evidence retrieval and relation estimates.
Future context (Lee et al., 2026)	Samples continuations and uses them as clues for black-box hallucination detection.	Generated continuations extend consistency checking without supplying independent external verification.
FaithLens (Si et al., 2026)	Trains detection and explanations using filtered synthetic data, supervised fine-tuning (SFT), and rule-based reinforcement learning.	Explanation quality and decision accuracy require separate validation; fluency alone does not establish explanation faithfulness.
QASemConsistency (Cattan et al., 2026)	Decomposes claims into predicate–argument QA pairs for fine-grained support localization.	Semantic error localization supplies a finer target than response-level binary classification.

Table 8: Additional hallucination detection and explanation methods for LLMs.

Study	Mechanism or evaluation design	Comparative implication and boundary
Factual confidence (Mahaut et al., 2024)	Compares confidence estimators and checks stability under meaning-preserving input changes.	Confidence can be unstable across paraphrases; access and supervision requirements differ.
New-knowledge fine-tuning (Gekhman et al., 2024)	Controlled closed-book QA experiments vary the amount of unfamiliar factual information in SFT.	The findings motivate knowledge-aware supervision within the studied fine-tuning setup.
Known facts (Jiang et al., 2024)	Examines layer-wise prediction dynamics when questions about the same fact yield correct versus incorrect answers.	Errors can reflect knowledge access rather than absence; measured patterns do not prove universal causal mechanisms.
Lost in the Middle (Liu et al., 2024c)	Evaluates how position of relevant evidence affects long-context use.	Evidence being present is insufficient for reliable use; its tasks are not themselves a comprehensive hallucination taxonomy.
Hidden-state limits (Servadio et al., 2025)	Tests factuality encoding on more realistic, model-dependent generated data.	Accuracy on synthetic true/false facts does not establish probe generalization to target-model outputs.
Unfamiliar supervision (Kang et al., 2025)	Relates hallucination response patterns to supervision of unfamiliar fine-tuning examples, including reward-model errors.	Factuality alignment depends on how unknowns and reward errors are labeled.
Illusion of Progress (Janiak et al., 2025)	Examines lexical-overlap labels and response-length heuristics in detector evaluation.	Semantic validation and simple baselines remain necessary; an LLM judge is not automatically a gold standard.
CoT obscures cues (Cheng et al., 2025)	Measures how reasoning prompts shift hidden/probability signals and detector effectiveness.	A generation intervention shifts the detector’s input distribution, making calibration transfer uncertain.
Hallucination tax (Song et al., 2025)	Studies refusal degradation after reinforcement fine-tuning, primarily on unanswerable mathematics, with reported transfer to factual QA.	Evidence is not a universal result for every factual-generation task.
Source preferences (Schuster et al., 2026)	Controlled synthetic-source study of source preferences and repetition in conflicting English evidence.	Frequency and source cues influence conflict resolution; synthetic-source preference is not proof of real-world source credibility.

Table 9: Studies of hallucination mechanisms and reliability in LLMs.

Study	Mechanism or evaluation design	Comparative implication and boundary
FaithfulRAG (Zhang et al., 2025a)	Models fact-level conflicts between retrieved and parametric knowledge before response generation.	Explicit reconciliation complements source preference; source adherence and world correctness remain distinct.
CoCoA (Khandelwal et al., 2025)	Adapts token-level decoding using confidence/context distribution differences.	Adaptive and fixed contrast differ in conflict sensitivity; low-conflict behavior also matters.
Context-DPO (Bi et al., 2025)	Constructs preferences from faithful versus stubborn responses under knowledge conflict.	A context-faithfulness objective differs from truthfulness with respect to external facts.

Table 10: Additional hallucination mitigation methods for LLMs.

D A Common Reporting Protocol

The following protocol is a synthesis of the comparisons in Sections 3–6, not a benchmark run. It can be used to evaluate whether an intervention in Figure 4 solves the kind of error illustrated in Figure 1 under the target in Figure 3.

D.1 Fix the target before selecting a metric

Specify whether the target is world correctness, source support, or both; the answer unit; and the treatment of unknown or unverifiable claims. Record the generator/version, prompts, retrieval corpus/date, evidence truncation, decoding parameters, and judge/version. Use the same evidence and question distribution across systems where possible. If a method changes retrieval, separate retrieval quality from generation conditional on the retrieved passages.

For a nonempty extracted claim set C and fixed evidence E , a schematic support precision is

$$P_{\text{support}}(C, E) = \frac{\sum_{c \in C} \mathbf{1}[E \models c]}{|C|}. \quad (3)$$

This expression describes an aggregation principle, not the complete implementation of every factuality metric. Extraction determines the denominator; duplicate, trivial, subjective, and undecidable claims need explicit handling. An empty answer is not assigned perfect useful factuality by this definition. FActScore and VeriScore motivate claim-level analysis, while Core highlights the role of informative subclaims (Min et al., 2023; Song et al., 2024; Jiang et al., 2025).

D.2 Separate detection, localization, and generation quality

For binary detection, report class prevalence, confusion counts, precision/recall/F1, and balanced accuracy, $\frac{1}{2}(\text{TPR} + \text{TNR})$, where appropriate. Fix thresholds on independent validation data. For

spans, specify tokenization and exact versus overlap matching. For continuous scores, report discrimination and calibration separately; neither establishes the truth of an individual answer. Inspect false-positive and false-negative examples across error types, evidence lengths, and generator families.

For generation, pair factuality with unique supported content and required-aspect coverage. A source-faithful answer may still omit the answer to the question. Human auditing should sample both high- and low-scoring outputs, use explicit instructions, and report disagreement or adjudication. LongEval provides relevant human-evaluation guidance (Krishna et al., 2023). Avoid pooling scalar scores whose claim extraction, evidence, or unknown-label policy differs.

D.3 Measure selection and revision effects

Let A_τ be the answers retained by threshold τ among N questions, and let e_i indicate an erroneous answer under the declared reference frame. For nonempty A_τ , define

$$\begin{aligned} \text{Coverage}(\tau) &= \frac{|A_\tau|}{N}, \\ \text{Risk}(\tau) &= \frac{\sum_{i \in A_\tau} e_i}{|A_\tau|}. \end{aligned} \quad (4)$$

Comparing risk at matched coverage prevents a system from appearing universally better merely by refusing more questions. This response-level selective-evaluation definition is distinct from SEAL’s token-level training objective. For revision, count corrected errors, newly introduced errors, retained supported claims, and required information lost. Measure both before and after revision on the same initial drafts.

D.4 Audit independence, robustness, and cost

A verifier used for training or candidate selection should not be the sole final judge. Use indepen-

Family / example	Required access or supervision	Principal extra resource	Important boundary
Source alignment / AlignScore	Source and output text; trained auxiliary model	Verifier inference over chunks	Auxiliary-model errors and evidence truncation
Claim verification / FActScore	Claim extraction, knowledge source, retrieval and verifier	Search and per-claim checks	Retrieval failure differs from disproof
Self-checking / Self-CheckGPT	Repeated outputs; no generator internals	Additional generation and comparisons	Repeated falsehood can appear consistent
Semantic entropy	Samples and semantic equivalence; likelihoods for weighted estimation	Sampling and clustering	Estimator access differs by variant
Probe / Lookback Lens	Attention maps and labeled training spans	Feature extraction and probe inference	Supervised contextual proxy, not proof of truth
Factuality alignment / FactAlign	Generator training and sentence-level factual feedback	Verification plus optimization	Reward errors can be reinforced
Retrieval control / Self-RAG	Training, retrieval, reflection-token supervision	Training and retrieved candidates	Reflection judgments need auditing
Contrastive decoding / CAD	Context-conditioned and context-free logits	Extra distribution at decoding	Greater source adherence can amplify false evidence
Layer contrast / DoLa	Intermediate layer predictions	Layer-output computation	Parametric knowledge is not external evidence
Representation editing / TruthX	Hidden states and learned auxiliary components	Auxiliary learning and intervention	Transfer beyond training conditions is uncertain
Verification / CoVe, CRITIC	Model calls; CRITIC also needs tools	Verification and revision rounds	Isolated prompts are not independent knowledge
Selective generation / SEAL	Rejection-token training and regularized decoding	Training; altered answering behavior	Lower risk must be interpreted at matched coverage

Table 11: Access requirements and computational resources of representative methods. Resource descriptions are qualitative and do not imply measured cost rankings.

dently collected labels or an independently audited evaluator. Test natural errors from held-out generators, difficult consistent falsehoods, evidence conflicts, context permutations, paraphrases, and updated facts. Disaggregate performance rather than interpreting an average as proof of transfer. Verify with Caution and temporal benchmark analysis motivate independent error inspection and timestamped evidence (Godbole and Jia, 2025; Jiang et al., 2026).

Table 11 summarizes method access and deployment trade-offs. Report offline annotation and training costs separately from inference calls, sampled tokens, retrieval requests, tool invocations, latency, and memory. A black-box method can be expensive, while an internal method can be unavailable even if computationally economical. Do not infer a fixed latency multiplier from the number of model distributions or stages without measurement. Comparisons should either match budgets or present a quality–cost trade-off.